

Shabbir Hussain

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Experience

Amazon, Seattle,

Senior Software Development Engineer

July-2022-Present

- Created AI agents using AWS Agent Core to fix book/author data mismatches and calculate US sales on Amazon.com
- Developed Kubernetes applications with GPU scheduling and custom operators, enhancing system scalability by 45%
- Built web service for deployment of YAML configs via Flux on EKS to provision VSCode containers for adhoc Spark analysis
- Designed code with reusable CDK construct which can deploy machine learning infrastructure in multiple AWS account.
- Mentored team members to improve their understanding in creating **scalable ETL** process using **Big Data** technologies.

Data Engineer-II

Sep-2019-June-2022

- Developed **end to end system** of Speed Impact Pipeline (SIP) to process training data and run econometric model on EMR
- Created code to parallelize winsorization of two-dimensional matrix within SIP model using **multithreading**
- Owned the **CPT** planning and AWS infrastructure cost of Promise science team and maintained IMR spending every month
- Migrated SQL queries from redshift to **spark** for AFN-FCP to improve the **runtime** by 30% and reduce **IMR** by 60%
- Designed **workflows** to gather training data from **Redshift**, **EMR** and **Rest API's** in S3 and run parallelly multiple models

comScore, Reston, Virginia

Sep-2017-Sep 2019

Software Engineer

- Designed complete **architecture** to integrate **Snap Chat** data in **ComScore** datasets under data governance norms
- Built a **Scala** library which can be consumed by multiple **Spark** jobs having similar methodology at Comscore
- Created system to integrate aperiodic DVR data with existing TV and OTT data pipeline to gather user **ad impressions**
- Implemented custom input formats to read data from 10K partitions to enhance the **efficiency** of Spark jobs by 35%
- Secured second position globally in **comScore hackathon** by creating spark job monitoring tool to analyze custom matrices for each job launched in cluster using **Graphite** and **Grafana**

Solid Practice, Sarasota, Florida

May 2016 - Jan 2017

Software Engineer Co-op

- Developed application to encrypt and decrypt HL7 message using **Java** and HAPI to automate flow between lab and EMR
- Implemented RESTful web service in **Spring MVC** for fetching and validating data between two different databases
- Programmed algorithm using **Collections** to map multiple patient related questions & answers for various lab tests

Tata Consultancy Services, Software Engineer

Oct 2013 - Aug 2015

- Designed web application for fetching data from different data sources using **C#, MVC5 and WCF** to compare ad-click campaigns of different media providers, which resulted in revenue increase of 25%
- Built a tool using **ASP.NET, Entity Framework** and **SQL Server** to automate excel log entries increasing efficiency by 70%

Education

Northeastern University, College of Engineering, MA

Master of Science in Information Systems

Sep 2015 - Aug 2017

Relevant Courses: Engineering Big Data Systems, Cloud Computing, Data Structure & Algorithms, Web Development Tools, Multithreading & Parallel Processing, Application Engineering & Development, Database Management & Design

Rajiv Gandhi Technical University

Bachelor of Engineering (B.E.), Electronic and Communication Engineering

Sep 2009 - May 2013

Technical Skills

Languages	: Java/J2EE, Scala, Python, C#, SQL
Aws Service	: EMR, EKS, Agent Core, EC2, GLUE, S3, Redshift, VPC, RDS
Big Data & Cloud	: Spark, PySpark, Pig, Hadoop, MapReduce, AWS EMR, Greenplum
Web Technologies	: Spring MVC, Hibernate, ASP.NET MVC5, RESTful Web Services, jQuery